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UNIVERSITI TEKNOLOGI MALAYSIA

SAIA 1013: Responsible AI and Ethics

MEETING 6
SECTION 01

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MEETING 6

Governance for Responsible AI

- Ethical Framework Associated with AI (Framework with those principles)
- Sustainable AI Ethics Framework & Methodology (Mapping to SDG and ESG)
- Principle of Global Community & Societal Responsibility (AI into CSR commitments)

(Some are in form of these activities: Lecture, Discussion, Case Studies, Short Quiz, Self Study)

Ensuring Responsible AI in Practice - Governance

"If you can change the world by innovation today so that you can satisfy more of your obligations tomorrow, you have a moral obligation to innovate today"

Start with the right "Intention" -> Do it ethically

*Where we consider our own responsibility as researchers, developers and users of AI systems and evaluate the societal instruments needed to ensure responsibility. Above all, h*ow to ensure all stakeholders *act responsibly*.

Responsible AI Can Also Mean:

- ✓ **Having Policies & Governance in place** – Regulations, guidelines, incentives, societal deployment.
- ✓ **Understanding Developers' Role** – Individual/collective responsibility, professional codes of conduct.
- ✓ **Considering Inclusion & Diversity** – Avoiding bias, ensuring fair access, broadening education.
- ✓ **Composing Narratives of AI** – Balancing optimism, risks, and avoiding exaggerated dystopian fears.

Issue: Putting all these under the same "Responsible AI" umbrella risks confusion → may fuel fear of AI instead of solving real problems.

Recap: AI Ethics, Responsible AI & AI Governance

WHY

AI Ethics : The moral compass



AI Ethics as the **values and principles** we use to decide what is **right or wrong** when creating or using AI.

Think Right

The needs of society & humanity. Its scope is much broader and not limited to individual organization, sector, or region.

WHAT

Responsible AI : The good practice



Responsible AI is about **putting those ethics into action** by design, develop and using AI in a way that is **fair, safe, explainable, inclusive**

Do Right

Typically centers the needs of individual organizations & technology developers. Its scope is limited to users of said technology, impacted stakeholders, and compliance with existing laws/regulations

HOW

AI Governance : The rules and regulator



AI Governance is the **system of rules, policies, risk and oversight** that accountable and ensures AI is developed and used **responsibly and ethically**

Make Sure Others Do Right

Requires looking at questions pertaining to the trustworthiness of AI Governance tools with a strong normative push.

Recap: AI Ethics, Responsible AI & AI Governance

Scenario: A company uses an AI system to automatically screen thousands of job applications.

WHY ethics matter in AI hiring

- Avoid discrimination based on gender, race, age, or background
- Respect human dignity, fairness, and equality
- Prevent harm and ensure just decision-making

WHAT we need to do and build to reflects ethical value

- Use diverse and representative training data
- Provide explainability: Why was a candidate rejected?
- Involve human oversight in sensitive decisions
- Test for bias, accuracy, and reliability regularly

HOW to enforce, monitor, and hold systems accountable.

- Create clear internal AI policies
- Follow national and international AI frameworks (e.g. Malaysia's AI-RMap, EU AI Act)
- Establish ethical review boards or audit systems
- Conduct regular reporting and compliance checks

AI ETHICS

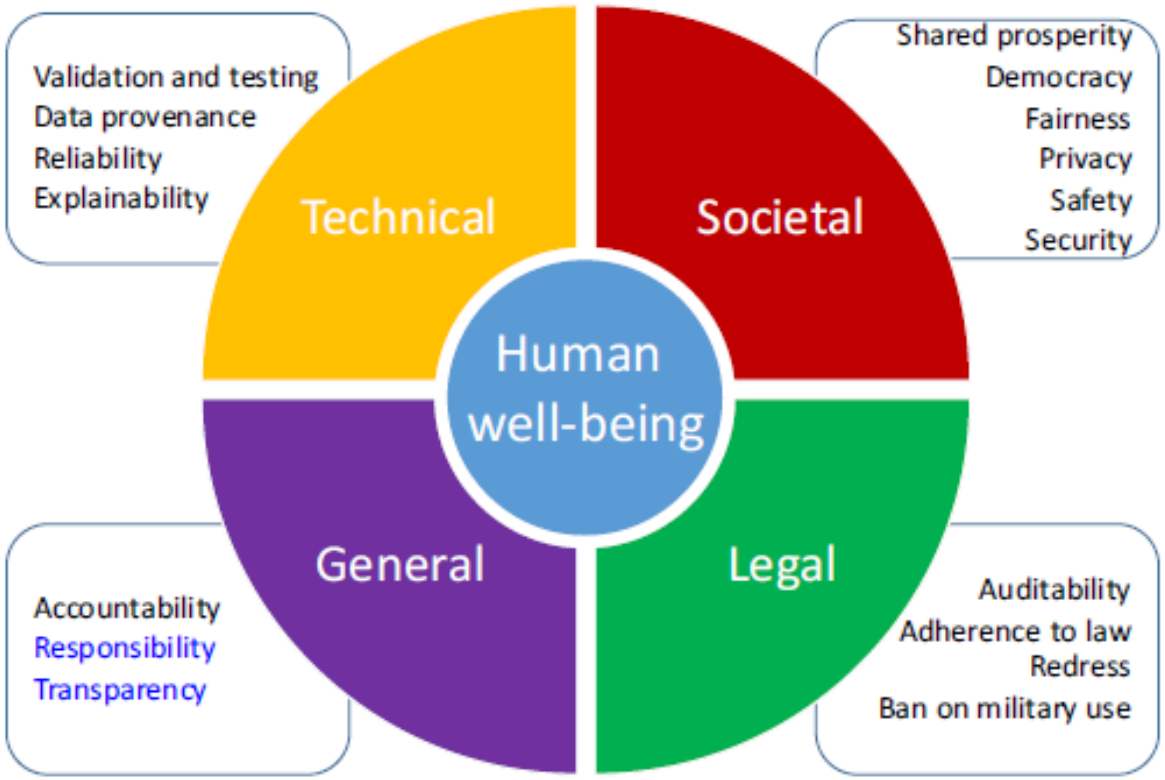
RESPONSIBLE AI

AI GOVERNANCE

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What to Govern in Responsible AI

- Human well-being → central priority.
- Accountability & Responsibility → broadly recognized.
- Principles fall into three main categories:
  - Societal – fairness, inclusion, human rights.
  - Legal – regulation, transparency, accountability.
  - Technical – safety, robustness, reliability.



Development of Governance for Responsible AI

- Growth of Governance Efforts
  - Increasing attention on ethical, societal, and legal impact of AI.
  - Initiatives come from:
    - ✓Top-down → national & transnational governance bodies (EU, OECD, UK, France, Canada).
    - ✓Bottom-up → practitioners, scientific community.
- Example of Early Major Governance & Policy Initiatives
  - IEEE initiative on Ethics of Autonomous and Intelligent Systems.
  - EU High-Level Expert Group on AI.
  - Partnership on AI (multi-stakeholder collaboration).
  - French “AI for Humanity” strategy.
  - All other governance based on different countries and nations
- **Goal:** → Provide **concrete recommendations, standards, and policies** to guide responsible AI development, deployment, and use.

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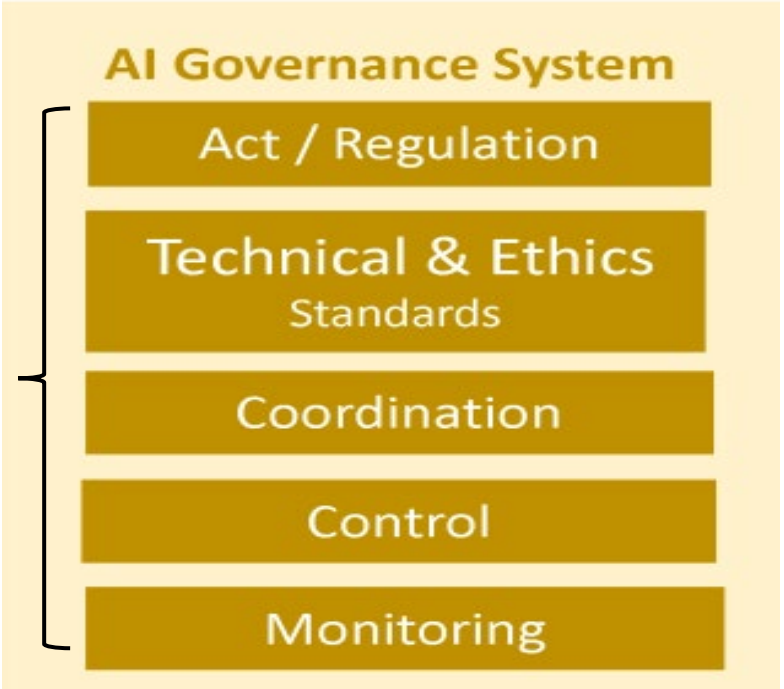
Governance for Responsible AI

Governance for Responsible AI
Turning principles into action through these TWO basic mechanisms:

- ✓ Regulation
- ✓ Certification

Dignum, V. (2019). *Responsible Artificial Intelligence: How to develop and use AI in a responsible way*. Springer

MALAYSIAN CONTEXT:
AI GOVERNANCE COMPONENTS



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ESTABLISHMENT OF AI GOVERNANCE SYSTEM



Establishing an AI governance system involves developing a structured framework that guides the ethical, legal, and responsible use of artificial intelligence technologies within your value chain.

By establishing your own AI Governance System related to your policy, you will gain a thorough understanding of AI technologies, their potential applications, and the ethical considerations associated with their use. This will also help you to develop a more comprehensive policy that details how AI technologies should be developed, deployed, and used within your value chain and include specific guidelines on data handling, algorithmic decision-making, transparency, bias mitigation, and accountability.

However, keep in mind that the governance system needs to implement a feedback loop and mechanisms for continuous improvement. Regularly reassess and adapt your AI governance system based on the evolving landscape of AI technologies, emerging ethical considerations, and lessons learned from previous projects.

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Act / Regulation

Ensure that your AI governance system aligns with existing legal frameworks and regulations. Address issues related to the 7 principles that may apply to AI applications in your jurisdiction.

Technical & Ethics Standards

Refine a set of 7 AI principles that will serve as the foundation for your policy. The standard should align with the 7 principles and the policy expectations.

Coordination

Clearly define roles and responsibilities for all stakeholders involved in the AI life cycle. Specify who is accountable for ethical considerations, compliance with policies, and oversight of AI projects that relate to the policy.

Control

Implement oversight mechanisms to ensure that AI systems adhere to established policies and ethical principles. This may involve regular audits, reviews, and assessments of AI projects to identify and address any ethical or legal concerns.

Monitoring

Integrate monitoring and evaluation processes into the AI governance system. Establish key performance indicators (KPIs) and metrics to assess the performance, impact, and ethical implications of AI systems throughout their life cycle.

Governance Mechanism for Responsible AI

Regulation is often seen as a “barrier” to innovation, but in AI, it is actually an enabler of trust and safety)

- Concerns: May stifle innovation, current laws may be inadequate
- Urgency: AI already impacts society → regulation cannot wait
- Targeted Approach: Domain-specific rules (e.g., healthcare, military) > generic laws
- Enabler Role: Regulation as incentives & funding to promote ethical AI
- Collaboration: AI experts + legal experts = practical, enforceable laws

Regulation as an enabler (can drive innovation)

- Push for explainability & sustainable data use
- May challenge deep learning dominance
- Requires cooperation among scientists, policymakers, developers, ethicists

Governance Mechanism for Responsible AI

Certification in AI is like a “trust label” (similar to organic food or ISO standards).

- Trust & Transparency: Certification helps the public and businesses know whether an AI system is safe, ethical, and reliable.
- Complement to Regulation: While regulation provides a baseline, certification allows differentiation — companies can show they go “beyond compliance.”
- Examples: Mention ongoing initiatives such as IEEE’s ECPAIS, the EU’s proposal for independent oversight bodies, and how auditing services (e.g., PwC, Accenture) already provide AI ethics checks.
- Summary: certification is essential for building public confidence in AI, similar to how we trust airplanes because they’re certified safe.

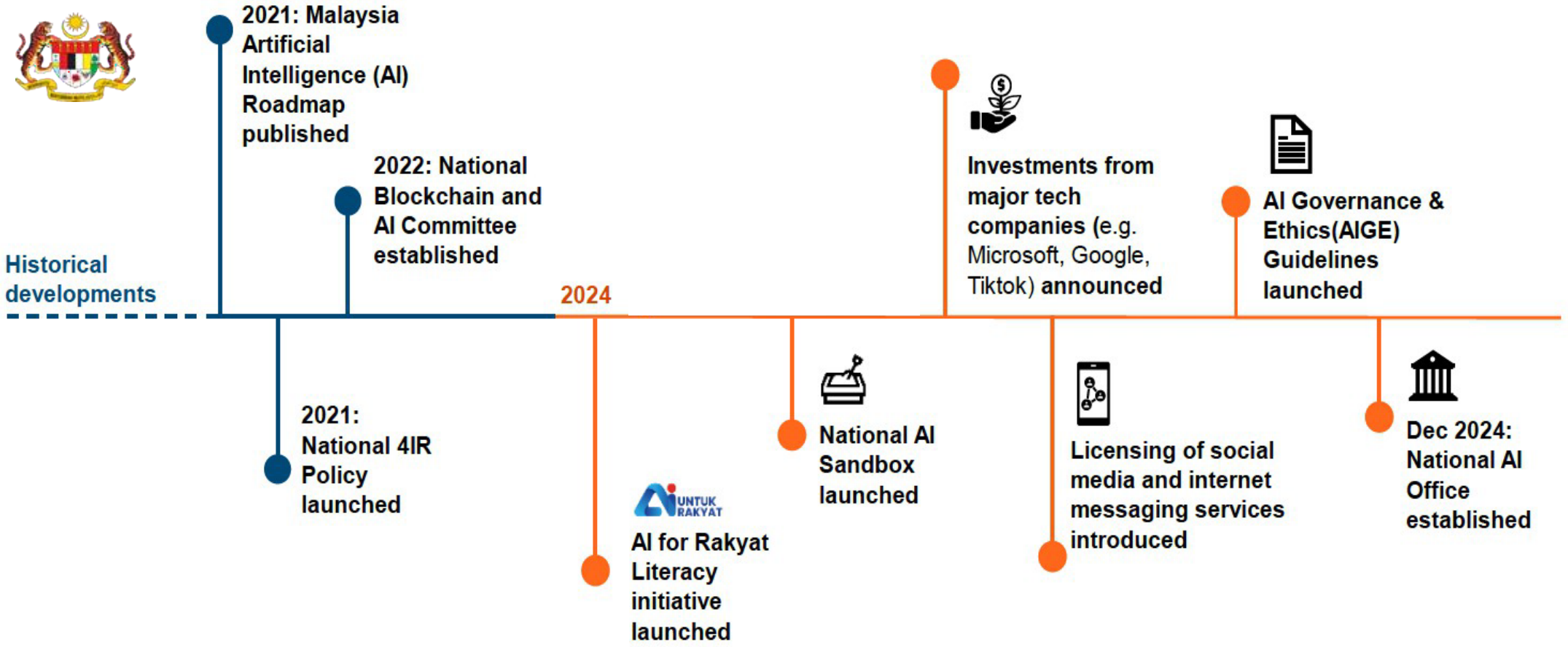
CUSTOMIZING RESPONSIBLE THROUGH AI CHECKLIST: EXAMPLE

Understandably, each policymaker's space and scope is unique. There is no specific formula that can cater to the stakeholders that you engage with. However, with an understanding of the AI Governance and Ethics Framework, including the life cycle and each component with its risk management, you could develop your checklist to cater to your own specific needs. Table 1: Example of a Policy Development Checklist related to AI in view with Cyber Security measures

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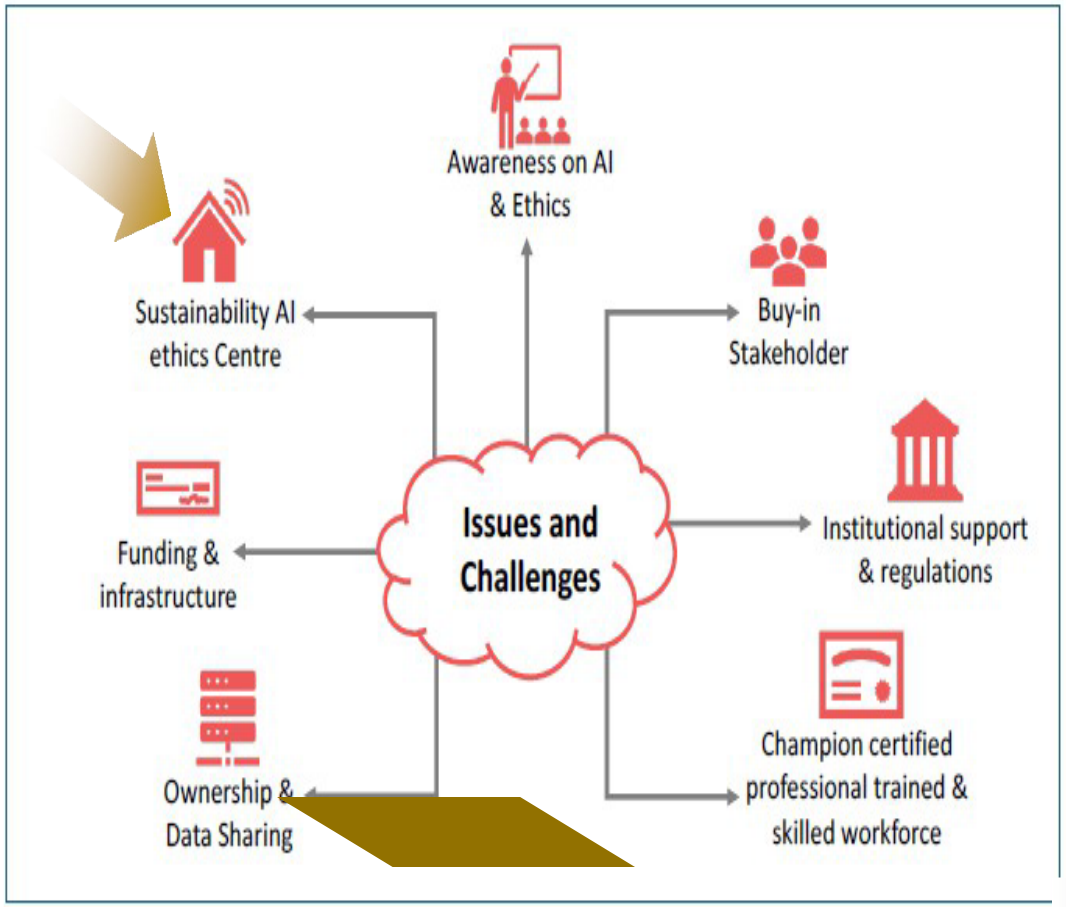
7 Principles	Description	✓
Fairness	• Require audits of AI training datasets to ensure they are fair, unbiased, and representative. • Implement algorithmic auditing procedures to regularly check AI systems for signs of unfair bias or discrimination. • Set up whistleblower protections and reporting channels for those who identify problems.	
Reliability, Safety & Control	• Set cybersecurity requirements like encryption, access controls, and vulnerability testing for high-risk AI systems. • Establish regulatory bodies to actively monitor the cybersecurity of critical infrastructure AI. • Require contingency plans and failsafes in case AI systems are compromised.	
Privacy & Security	• Set cybersecurity requirements like encryption, access controls, and vulnerability testing for high-risk AI systems. • Establish regulatory bodies to actively monitor the cybersecurity of critical infrastructure AI. • Require contingency plans and failsafes in case AI systems are compromised.	
Inclusiveness	• Fund education programmes on AI and cybersecurity skills for underrepresented communities. • Subsidise access to AI technologies for disadvantaged demographic groups and regions.	
Transparency	• Mandate documentation about an AI system's architecture, data sources, security protections etc. • Set up frameworks for external cybersecurity researchers to conduct audits of an AI system's codebase as needed.	
Accountability	• Require detailed cybersecurity audit logs showing who have accessed AI systems and what changes have been made. • Implement chain-of-custody cybersecurity tracking regarding AI data usage and algorithmic development.	
The Pursuit of Human benefit and Happiness	• Require the consideration of the appropriate level of human control for the particular AI system to ensure human benefits and happiness • Require the formation of a self-audit, response mechanism and an autonomous system to determine human rights and well being	

AI DEPLOYMENT AND ADAPTATION IN MALAYSIA



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ISSUES & CHALLENGES ON AI GOVERNANCE



Every sectors are at different degrees of digital transformation. A one-size-fits-all approach cannot effectively turn aspirations into actions.

Major digital divides remain within and across sector (rural/urban, connected/ unconnected).

Where plans and initiatives exist, implementation falters due to uncoordinated efforts and/or unclear mandates.

AI is still largely seen and used as a tool to maximize productivity and efficiency, rather than a tool to solve a wide range of complex challenges (urbanization, climate change, health outcomes, social equality, women's empowerment, economic diversification, digital literacy, etc.).

Sustainable AI Ethics

- Guidelines are tool for sustainable AI Ethics (concerns about sustainability is apparent)
- Can AI becomes an enabler to sustainability (positive of negative impact?)
- What is Sustainability? Sustainability are viewed from two perspectives:
  - ✓ SDG
  - ✓ ESG

MALAYSIAN CONTEXT:

THE NATIONAL GUIDELINES ON AI GOVERNANCE & ETHICS

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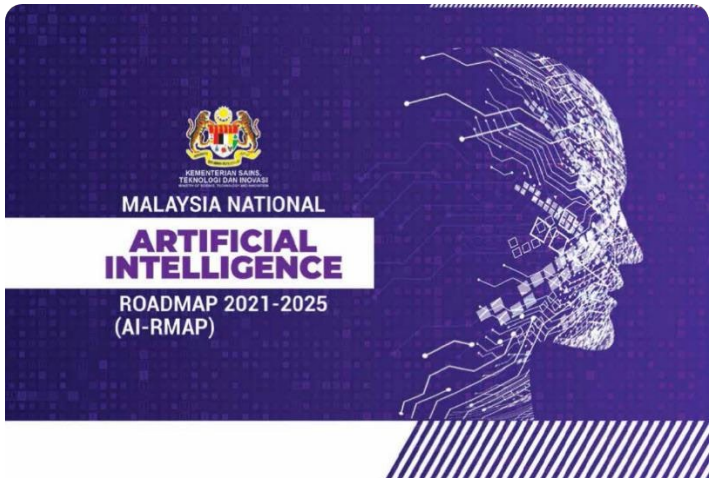
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AI DEPLOYMENT AND ADAPTATION IN MALAYSIA

National Artificial Intelligence (AI) Roadmap 2021–2025 (AUG 2022)

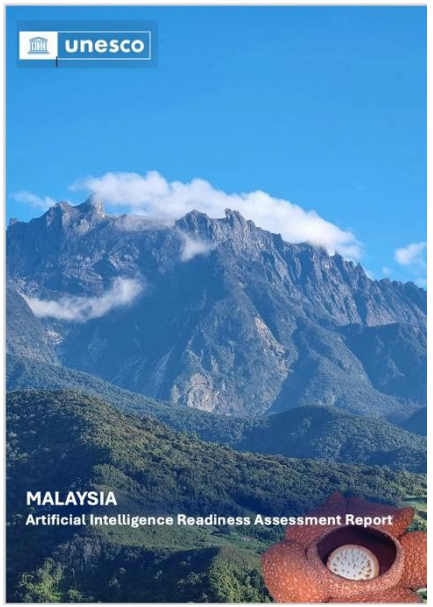


STRATEGIES	SUB-GOALS
01 Establishing AI Governance	AI Coordination and Implementation Unit (AI-CIU)
02 Advancing AI R&D	A Robust AI R&D Ecosystem
03 Escalating Digital Infrastructure to Enable AI	Digital Infrastructure for AI
04 Fostering AI Talents	AI Talents and AI Skilled Workforce
05 Acculturating AI	Increased AI Awareness and AI Adoption
06 Kick-Starting a National AI Innovation Ecosystem	AI Innovation Hub

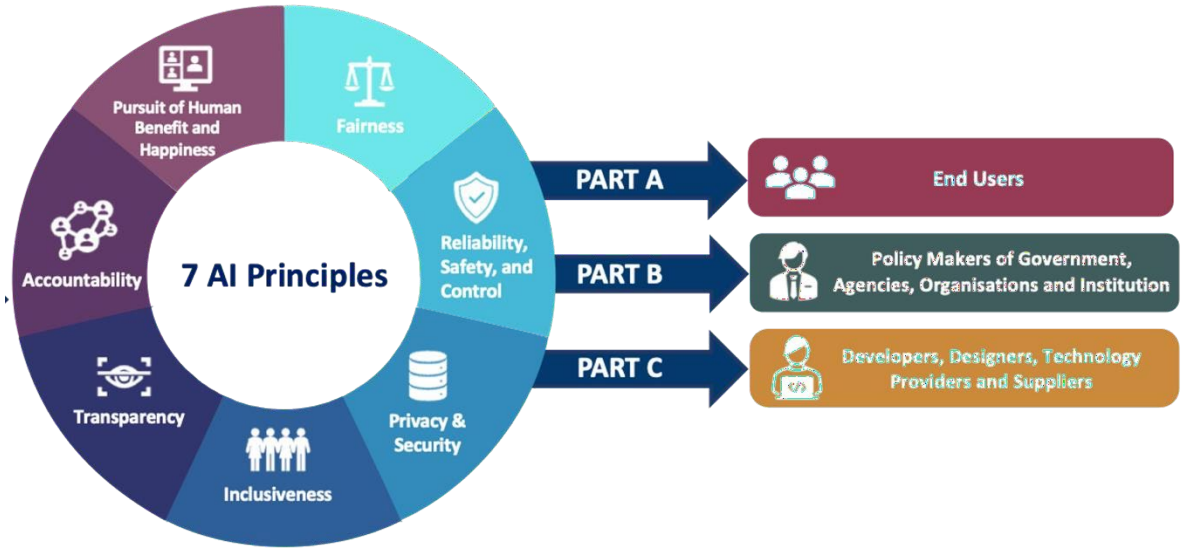
National Priority Area:

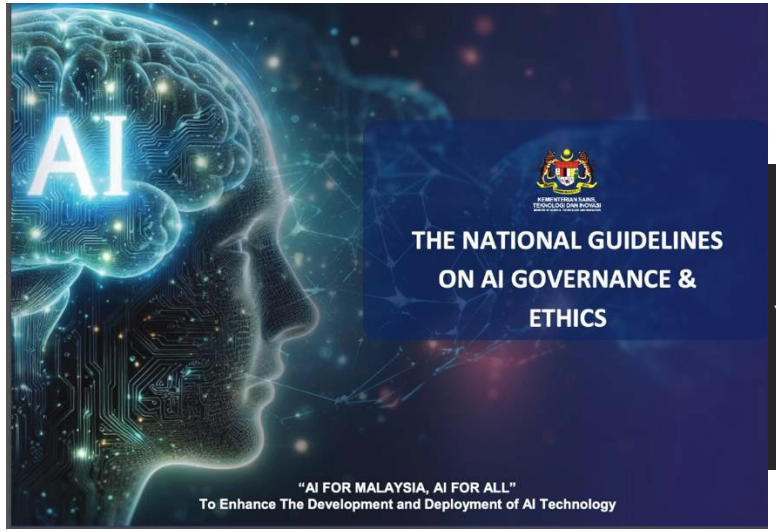
1. Agriculture & Forestry
2. Medical & Healthcare
3. Smart Cities & Transportation
4. Education
5. Public Service

Draft 2025



The National Guidelines AIGE (Sept 2024)





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National Guidelines Scope

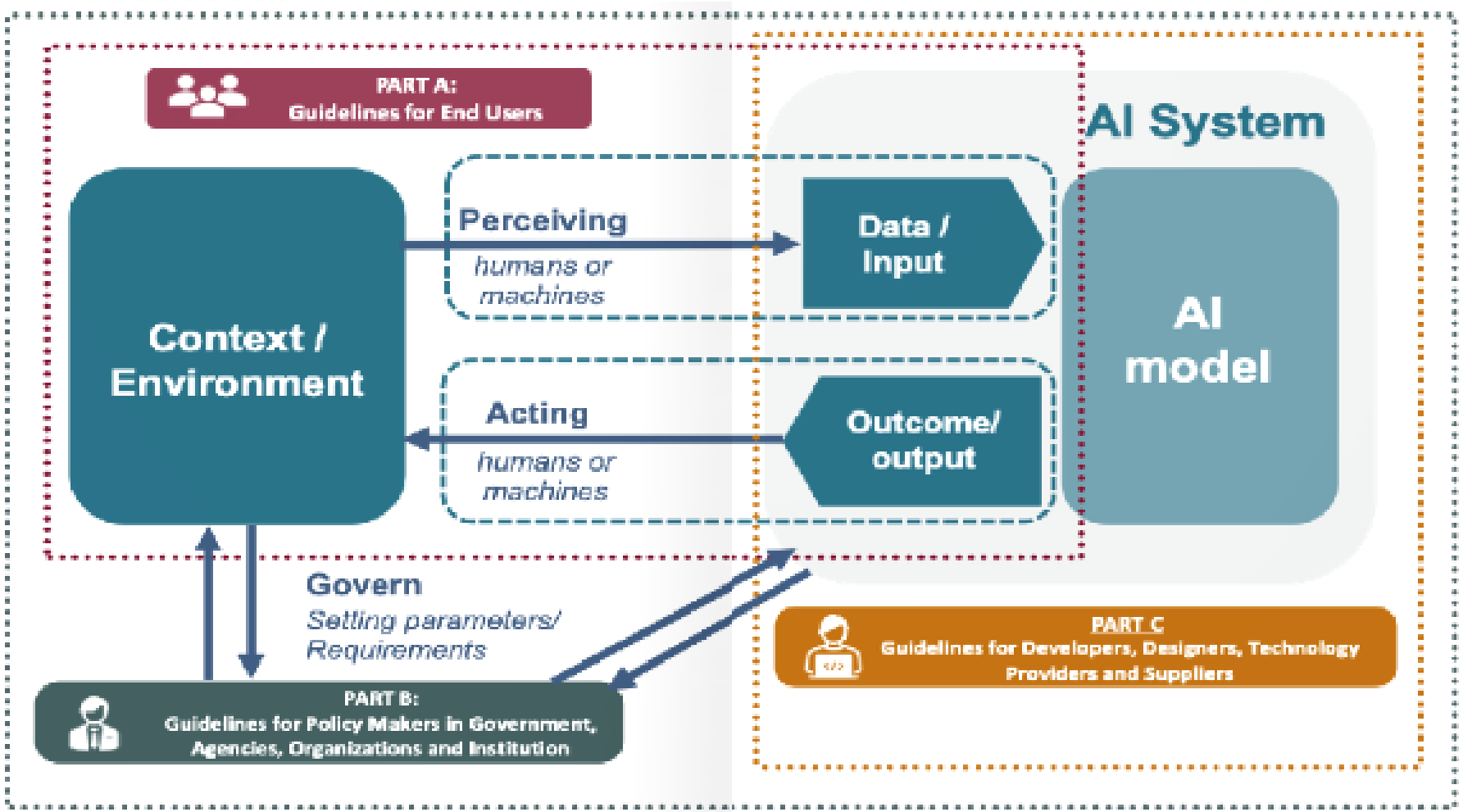


Figure 3: The scope of stakeholder and national guidelines covering

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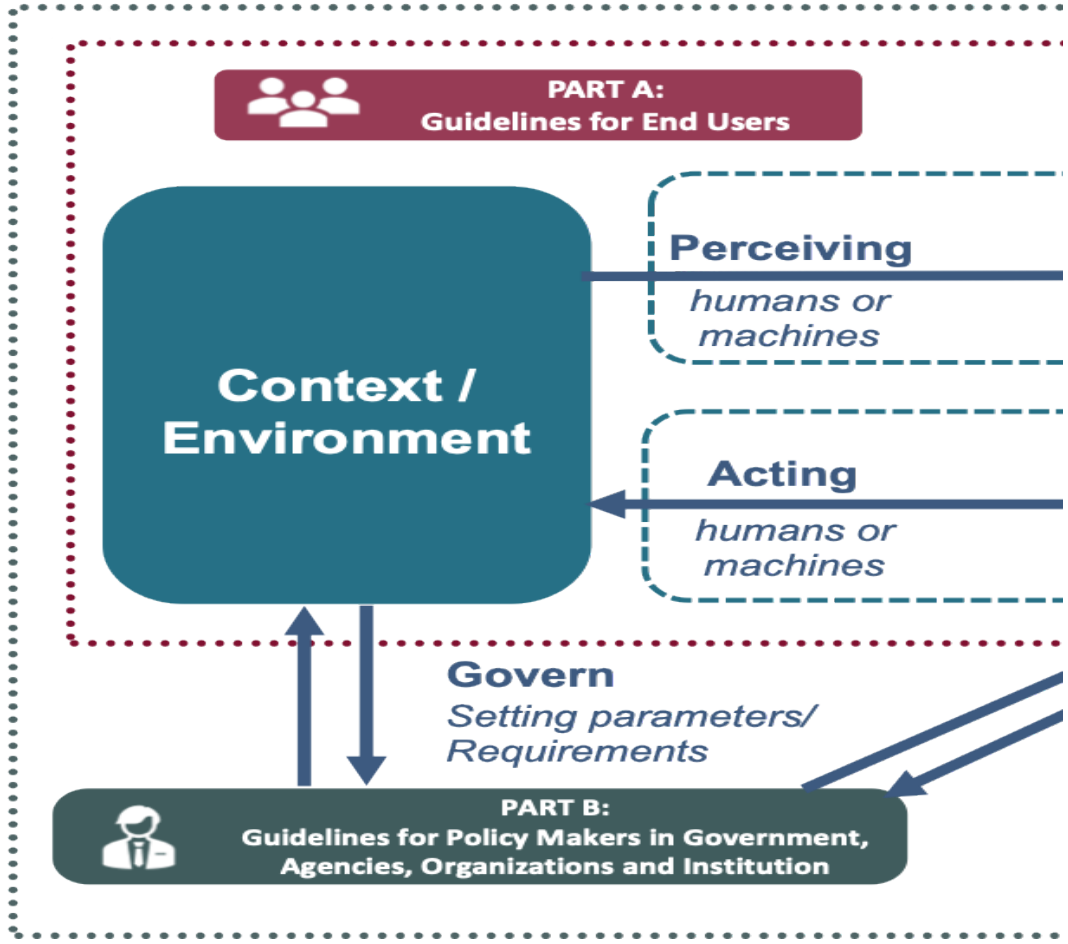
DESCRIPTION OF PART A, B, AND C

AI System as Structure for National Guidelines Development

The OECD defines AI systems as machine-based systems that, for explicit or implicit objectives, generate outputs like predictions, content, recommendations, or decisions, varying in autonomy and adaptiveness post-deployment.

These national guidelines cover stakeholders within the systems and include policy makers to govern within the system components. Stakeholder scope and national guidelines for stakeholders (End Users, Policy Markers, Developers) are illustrated in Figure 3. While these three stakeholder groups may not be exhaustive, they are necessary and fairly sufficient to establish the fundamental building blocks of an AI system. With this, a clear shared responsibility framework is developed to detail different roles and responsibilities among the key stakeholders. The shared responsibility framework needs to ensure that there is a clear delineation of roles and responsibilities between groups of stakeholders. It is important to note that, in reality, a single entity or person can certainly play more than one role.

For example, a specific stakeholder can hold multiple roles at the same time, depending on the particular use of AI. These categories of stakeholders can be mutually exclusive and collectively exhaustive. Each stakeholder can refer to parts A, B, or C according to their interest and requirements.



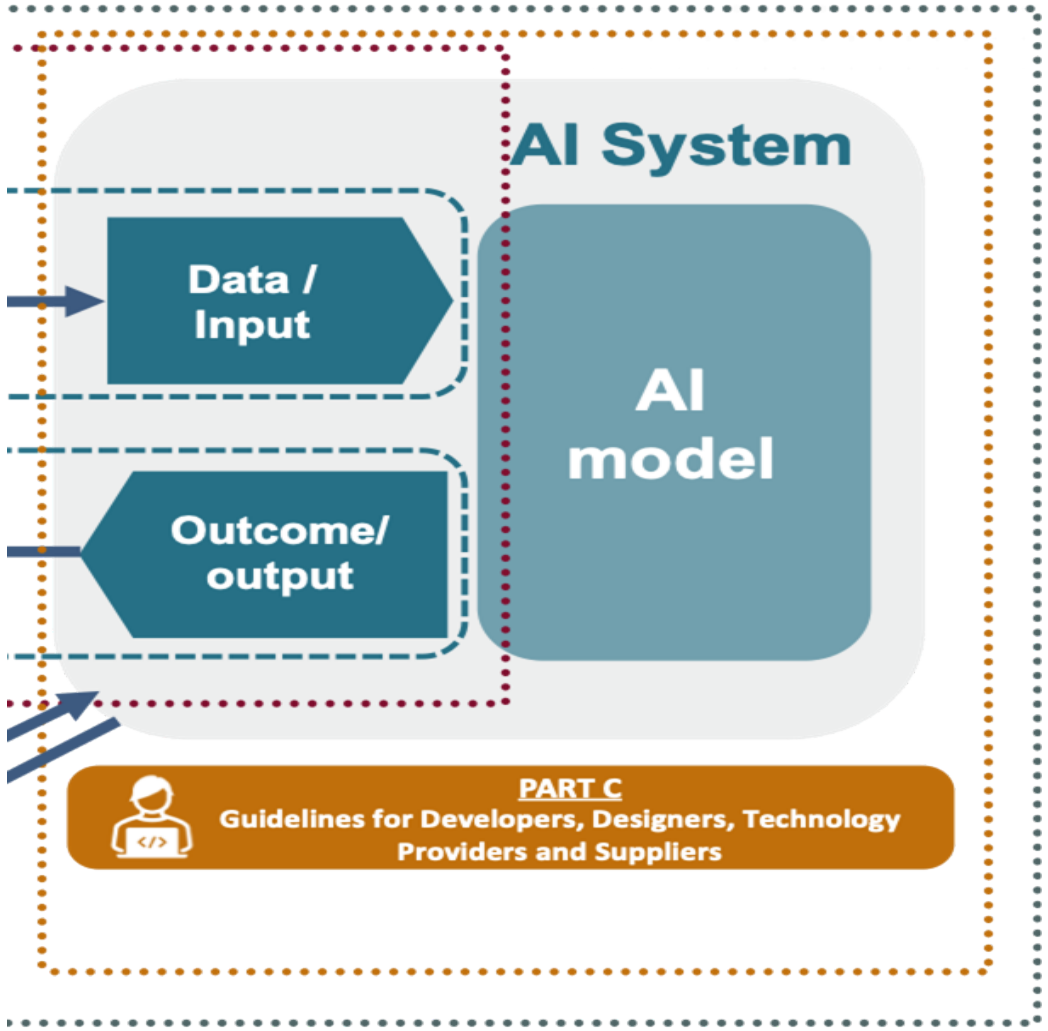
PART A End Users

End users are individuals or organizations who use AI products. Different groups of end users use AI products and services in different ways. Some use AI-powered virtual assistants, smart home devices, or AI applications on their smartphones.

Others working in different agencies and industries use AI-generated applications: government agencies use AI for security, law enforcement, or public services; manufacturers use it for quality control and to optimize their supply chain; banking institutions rely on AI for fraud detection; and content creators use AI for video editing or recommendation systems. Since these consumers may or may not be aware that they are adopting AI technologies, these guidelines may raise their awareness of responsible AI use, inform people about their rights and responsibilities, and empower them to make informed decisions regarding AI technologies and applications.

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DESCRIPTION OF PART A, B, AND C



PART B

Policy Makers of Government, Agencies, Organizations and Institutions

The primary target audiences for the National Guidelines are the policymakers, planners, and managers responsible for AI workforce policy and planning at national and local levels. Government officials, policymakers, and regulatory agencies are individuals or agencies that are responsible for formulating and implementing policies and regulations concerning the use of AI, and for ensuring ethical and Responsible AI practices across different sectors.

The National Guidelines provide a framework and practical recommendations to ensure the responsible and ethical use of AI within the jurisdictions of those responsible and guide them in ensuring compliance, consumer protection, and fair competition in AI-related activities.

PART C

Developers, Designers, Technology Providers and Suppliers

Developers and designers are those involved in developing and designing AI products. They come from different fields of expertise, such as computer science, data science, engineering, and AI research, and they work for different AI users and industries. The National Guidelines outline best practices, ethical considerations, and technical standards for Responsible AI development and deployment. The Guidelines also enable these professionals to align their work with recognized benchmarks for better outcomes and to reduce or mitigate any AI ethical risks.

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ALIGNING GOVERNMENT INITIATIVE WITH RESPONSIBLE AI AND SDG & ESG PRINCIPLES



SUSTAINABILITY PILLARS IN POLICY / GUIDELINE MAKING

The concept of sustainability is composed of three pillars: economic (Governance), environmental, and social.

- **Three-Pillar Balance**: Sustainable AI policy requires balancing economic, social, and environmental dimensions.
- **Responsible AI Principles** – Integrating responsibility in AI development and deployment ensures ethical and accountable use.
- **Long-Term Contribution** – This approach maximizes societal well-being and prosperity while minimizing risks and negative impacts.

For policymakers developing policies related to AI, sustainability encompasses several key dimensions which include:

- ✓ **Environmental Impact: Taking account of the environmental sustainability of AI technologies involves assessing the energy consumption and carbon footprint associated with AI hardware infrastructure, data centers, and computing resources. Sustainable AI policies may encourage the development and adoption of energy- efficient algorithms, hardware, and practices to mitigate environmental impact.**
- ✓ **Social and Economic Impact: Focusing on the social and economic sustainability of AI involves addressing potential societal system disruptions as well as its improvement. Sustainable AI policies aim to promote inclusive growth, create new job opportunities, and support workers in transitioning to new roles through education, training, and re-skilling programs.**

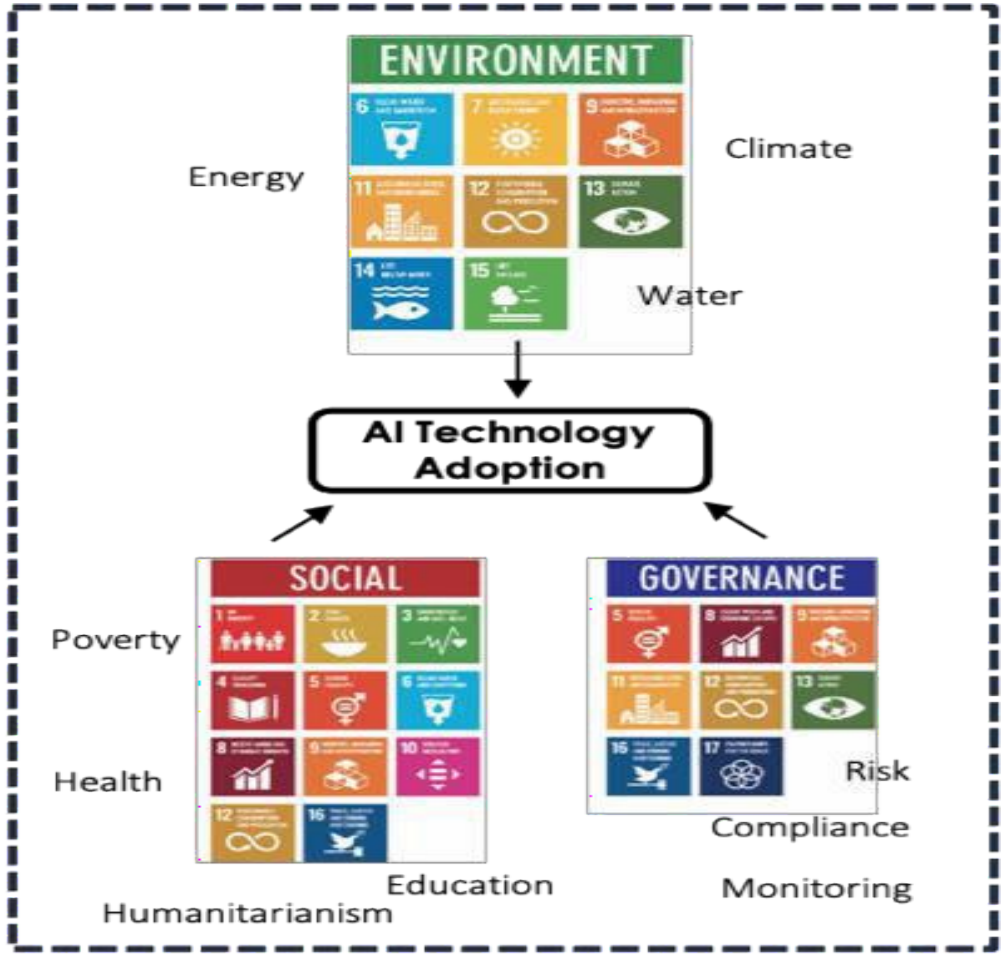
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ALIGNING INDUSTRIAL ECONOMIC CONTRIBUTIONS WITH RESPONSIBLE AI AND SDG & ESG PRINCIPLES

- **AI as a Driver of Sustainability:** AI can significantly support sustainable development and the achievement of the UN Sustainable Development Goals (SDGs) when designed and applied responsibly.

- **Importance of ESG Compliance**: Environmental, Social, and Governance (ESG) principles have become essential in addressing climate risks and ensuring companies account for their broader environmental and social impacts.

- **Role of ESG Data**: ESG data provides critical insights into sustainability performance, covering environmental (e.g., carbon emissions, water use), social (e.g., diversity, equity, human rights), and governance (e.g., anti-corruption, labor practices, board composition) dimensions.

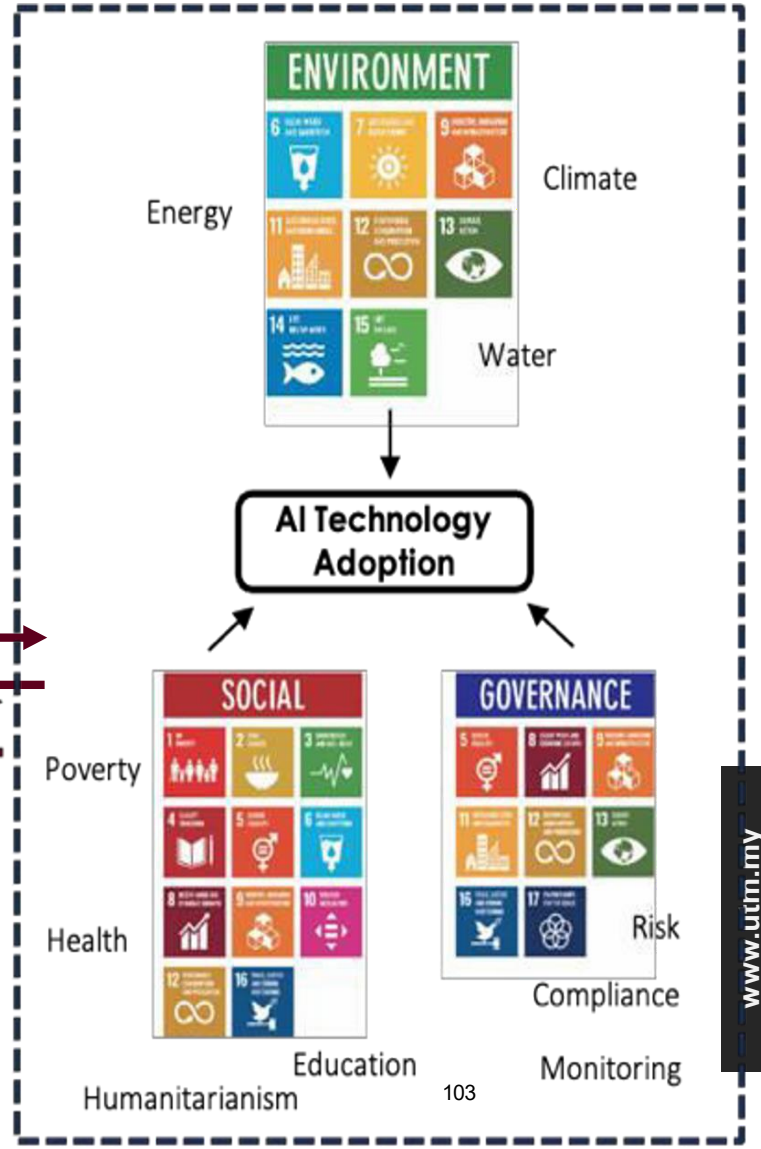


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ALIGNING INDUSTRIAL ECONOMIC CONTRIBUTIONS WITH RESPONSIBLE AI AND SDG & ESG PRINCIPLES

Examples of Context (Not limited to) :





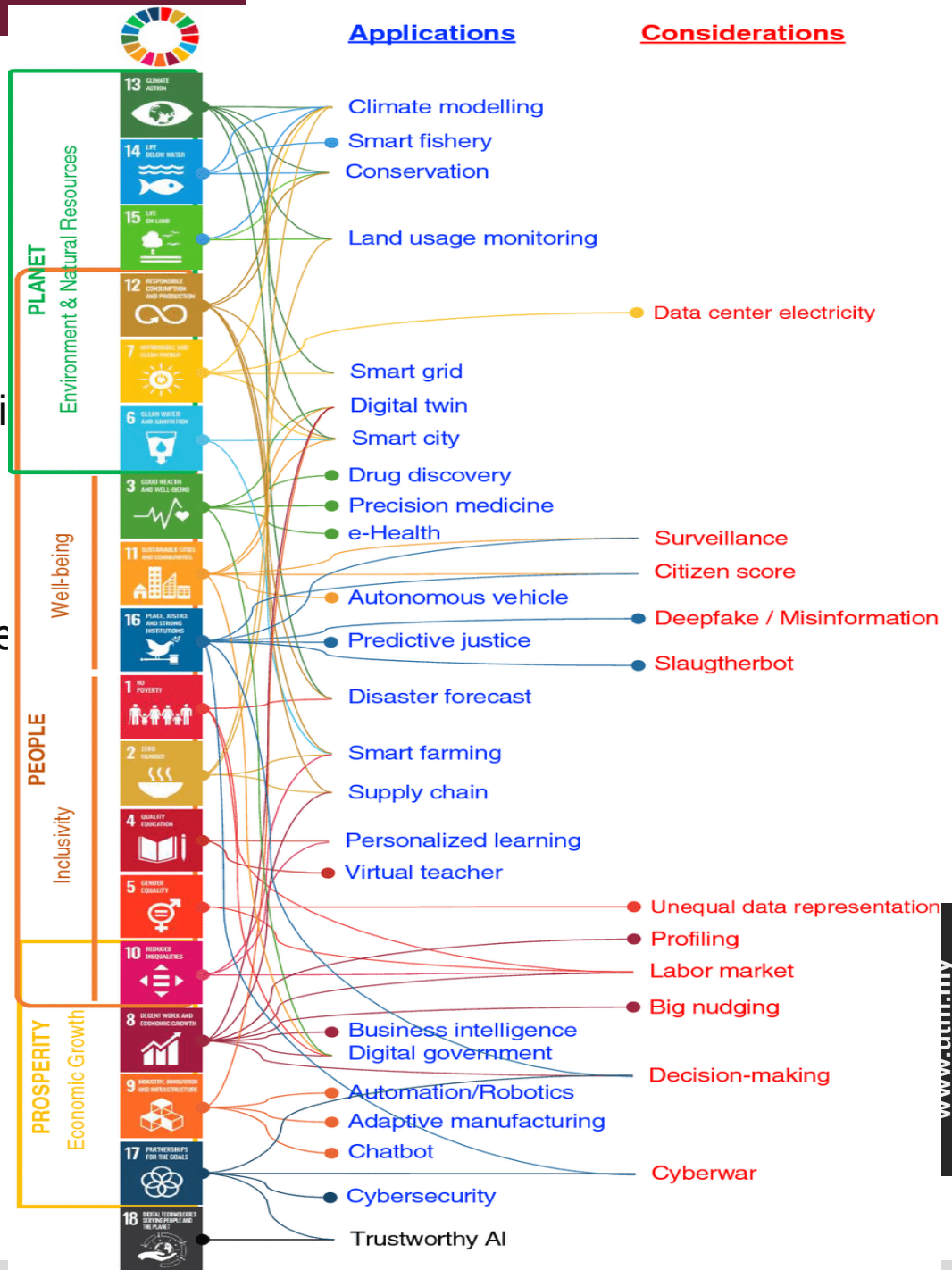
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AI Technology Adoption to Support SDG & ESG Reporting

01 Environmental data should capture environmental information such as annual carbon emissions and energy consumption, water usage, waste and polluti output, and more

02 Social data focuses on statistics related to workforce diversity, gender equity, human rights, and more.

03 Governance data racks company input regarding corruption, labor practices, gender composition of the board of directors, and more.



https://www.researchgate.net/figure/Applications-and-considerations-of-AI-in-relation-to-the-17-Sustainable-Development_fig1_356287044

AI Technology Adoption to Support SDG & ESG Reporting

Policy Development:

The research project's findings and recommendations will inform the development of policies and regulations related to ESG reporting and AI adoption in Malaysia, facilitating the creation of a supportive ecosystem for sustainable business practices

Enhanced ESG Reporting:

Companies will have improved capabilities to report their ESG performance, leading to better transparency, accountability, and informed decision-making.

Sustainable Business Practices:

The project will promote the adoption of sustainable business practices among companies, contributing to the achievement of the United Nations Sustainable Development Goals (SDGs) in Malaysia and the region.

Investor Confidence:

Reliable and comprehensive ESG disclosures will enhance investor confidence, attracting sustainable investments and fostering economic growth.

AI Technology Adoption to Support SDG & ESG Reporting

AI + ESG = Leveraging AI-Driven Solutions for ESG

Environmental (E):
Focus on impact of an organization's activities on the natural environment

AI application

Examples
- Optimizing Energy efficiency
- Enhancing renewable energy generation and distribution

Social (S):
Focus on an organization's impact on society, including employees, communities, and broader stakeholder groups.

AI-driven solutions

- Enhancing disaster response and resilience
- Promoting ethical and responsible AI development

Governance (G):
Focus on the systems, processes, and structures that guide an organization's decision-making and accountability.

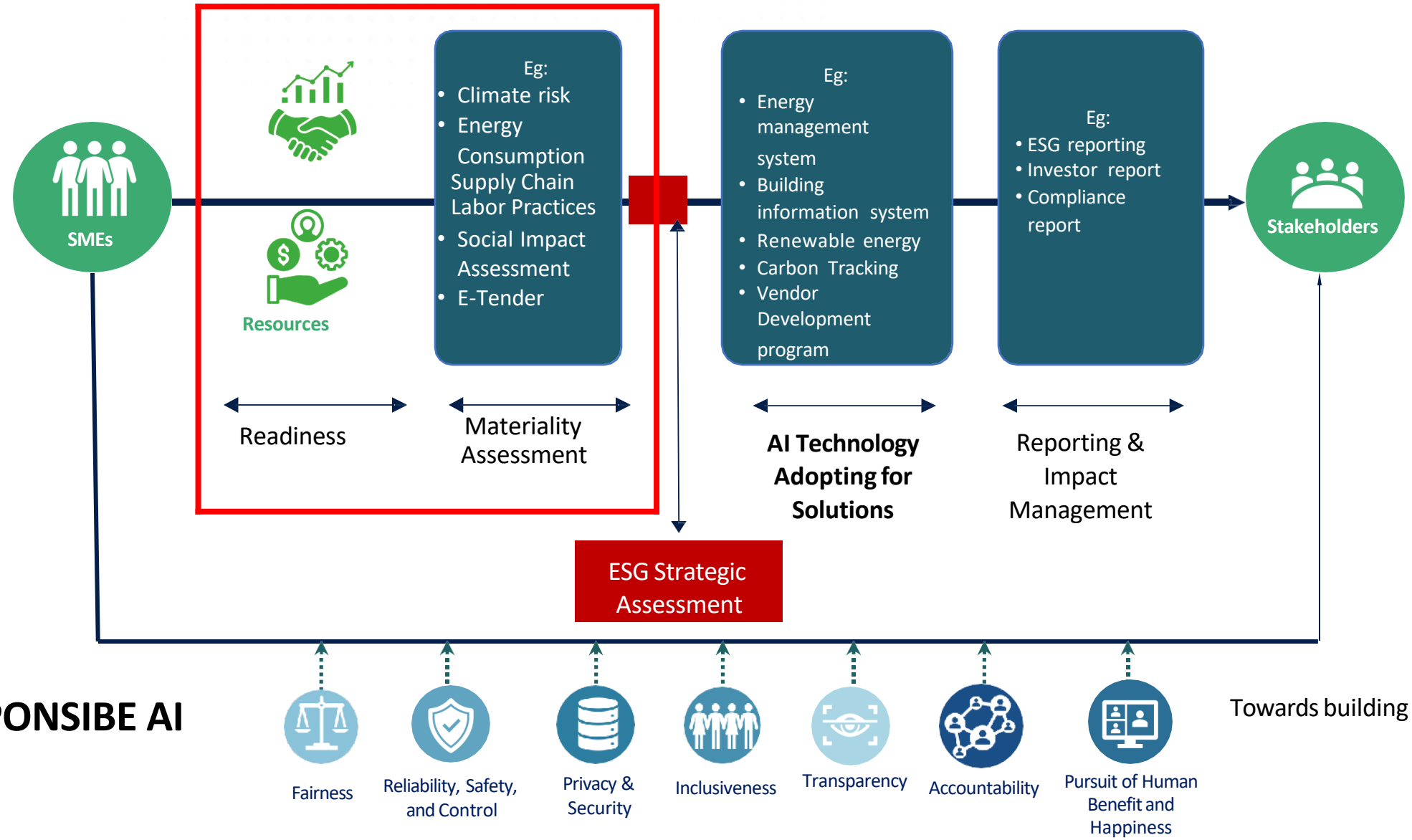
AI tech adoption

- Enhancing corporate governance and board diversity
- Advancing circular economy practices
- Enhancing ESG data quality and reliability
- Enhancing climate finance and investment decisions

Augmenting ESG reporting capabilities with AI support business to Sustain :

- competitive
- compliant
- aligned with stakeholders interest

ESG Strategic Assessment Process



Ethical Challenges & Accountability in AI-Driven ESG Analysis

Key Ethical Challenges

AI applied in ESG for data analysis, prediction, and scoring raises new ethical challenges and accountability questions.

Data Quality & Bias

- ESG datasets often lack standardization across regions and sectors.
- AI systems may amplify existing biases (e.g., favoring large corporations with better reporting mechanism)

ESG Scoring Algorithms

- Proprietary models are often black-boxes with limited explainability.
- Stakeholders may not understand or challenge automated ESG ratings.

Greenwashing Risks

- Companies may selectively disclose data to manipulate AI-driven ESG ratings.
- Lack of validation may let misleading ESG performance go unchecked.

Exclusion of Local or Community Metrics

- ESG-AI tools may prioritize globally recognized indicators and ignore local context, causing social exclusion or misrepresentation.

Ethical Challenges & Accountability in AI-Driven ESG Analysis

Accountability Measures & Governance

AI applied in ESG for data analysis, prediction, and scoring raises new ethical challenges and accountability questions.

Ethical AI Audits

Conduct independent algorithmic audits focused on fairness, bias, and explainability.

Transparent Methodologies

Publicly disclose ESG scoring logic and weighting schemes to allow stakeholder scrutiny.

Participatory Design

Engage diverse stakeholders, including civil society, in system design and performance evaluation.

Regulatory Alignment

Ensure AI-ESG tools comply with evolving standards (e.g., EU AI Act, OECD ESG Disclosure Framework).

Direct and Indirect SDG/ESG Impacts of AI Compute and Applications

Direct environmental impacts AI compute resources lifecycle

Production	Transport	Operations	End-of-life
• Raw material extraction • Assembly • Manufacturing	• Distribution • Freight transportation • Handling & storage	• Energy consumption • Water consumption • Carbon footprint	• Collection & shipping • Dismantling & recycling • Waste disposal

Indirect environmental impacts AI applications

Positive impacts	Negative impacts
• Beneficial sectoral applications • Climate mitigation and adaptation • Environmental modelling and forecasting	• Harmful sectoral applications • Carbon leakage (net increase in emissions) • Consumption patterns and rebound effects

Source: OECD.AI Expert Group on AI Compute and Climate, literature review, expert interviews

https://www.oecd.org/en/publications/measuring-the-environmental-impacts-of-artificial-intelligence-compute-and-applications_7babf571-en.html

Positive Indirect SDG / ESG Impacts of AI Application

01

AI becomes a powerful tool that drive and contribute to the achievement of SDGs while aligning with Environmental, Social, and Governance (ESG) principles.

02

ESG compliance has become increasingly important in the face of increasing climate risks as it deals with a company's impact on the environment and society.

03

ESG data includes any indicators that shed light into the sustainability context of an asset, facility, company or region, whether historic, current or expected.

Negative Indirect SDG/ ESG Impacts of AI Application

GHG Emissions

While AI can increase efficiencies that support sustainability efforts, these applications can also work to increase net GHG emissions instead of lowering them (eg: energy efficient facilities – data center excess heat)

Consumption Pattern & Rebound effects

Efficiency gains through technological progress are offset by acceleration in resource consumption (eg: energy-efficient AI applications)

Harmful Sectoral Application

Negative environmental impacts of the mining, extractive and manufacturing sectors, where advanced AI applications can be used upstream for finding and extracting minerals or fossil fuels, midstream for transport and material storage, and downstream for product refining

https://www.oecd.org/en/publications/measuring-the-environmental-impacts-of-artificial-intelligence-compute-and-applications_7babf571-en.html

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CONCLUSION ON AI GOVERNANCE AND ETHICS GUIDELINES

Why Guideline and Governance Matters:

- Ensure AI aligns with societal values, laws, and ethics.
- Build trust by prioritizing user well-being, safety, and transparency.
- Provide recourse when AI acts unethically or harmfully.

For Stakeholder:

- Consumers: Right to information on data use, decisions, and third-party sharing.
- Developers: Clear principles to avoid bias, discrimination, and harm.
- Policymakers: Frameworks to regulate risks and foster responsible innovation.

Broader Impact:

- Strengthen reputation & competitiveness through ethical practices.
- Promote industry standards, consistency, and interoperability.
- Drive sustainability, support SDGs, and align with ESG principles.

Last (3rd) Forum Participation (2%)

Select one AI, be it the existing one or future to be (under development). Briefly, identify 2 impacts(one positive and one negative) and relate them to the long term sustainable goal SDG / ESG. Match it to which SDG and under which ESG.



https://proindia.net/sustainable-development-aligning-esg-with-un-sdgs/



https://www.worldwidegeneration.co/news/from-esg-to-sdg

www.utm.my

VERY COMMON ANSWER.

Selected AI: *ChatGPT (Generative AI for education and productivity).*

Positive Impact:

- *Improved access to education* → AI tools like ChatGPT make learning resources widely available, reducing barriers for underprivileged groups.
- Linked SDG: SDG 4 – Quality Education.
- Linked ESG: *Social* – supports inclusivity and human capital development.

Negative Impact:

- *Risk of misinformation and bias* → AI can generate inaccurate or biased content, potentially harming trust and spreading false narratives.
- Linked SDG: SDG 16 – Peace, Justice, and Strong Institutions (promoting reliable information and trust in institutions).
- Linked ESG: *Governance* – highlights the need for stronger oversight, transparency, and accountability in AI development.

Conclusion:

The long-term impact of AI like ChatGPT depends on whether governance frameworks can minimize risks (misinformation, bias) while maximizing positive contributions to education and inclusiveness.